WEEK 4: DATA VISUALIZATION & CAPSTONE PROJECT

End-to-End Sales Data Analysis Case Study

1. Project Overview

This project analyzes a sales dataset using Python, Pandas, Matplotlib, scikit-learn, and Microsoft Power BI. The objectives were to explore the dataset, summarize sales and profit patterns, build a regression model to estimate sales, and create a dashboard for communicating business metrics.

2. Dataset Description

The supplied Excel dataset contains 700 rows and 16 columns. It includes information about business segment, country, product, discount band, units sold, manufacturing price, sale price, gross sales, discounts, sales, cost of goods sold (COGS), profit, date, month number, month name, and year. The records cover 2013 and 2014, with five countries, six products, and five segments.

Data-quality checks found 53 missing values in the Discount Band column and no duplicate rows. The Date column had 31 missing values. These missing values should be considered when interpreting date-based results; they were not silently assumed to be complete.

3. Exploratory Data Analysis (EDA)

Python and Pandas were used to inspect the data, calculate summary statistics, and examine relationships between numerical variables. Matplotlib visualizations included distributions, a correlation heatmap, monthly sales trends, and yearly sales totals. The Power BI report presents comparisons of sales by country, sales by product, profit by segment, and monthly sales over time.

These views help stakeholders compare business performance across countries and products, examine profit by segment, and look for changes in sales over time. The dashboard is intended to support exploration and discussion rather than establish causes for the observed patterns.

4. Sales Prediction Model

A Linear Regression model was trained to predict the Sales column using Units Sold, Manufacturing Price, Sale Price, Discounts, and COGS. The data was split into 560 training rows (80%) and 140 testing rows (20%), using random_state=42 for reproducibility.

Evaluation metric	Result
Mean Absolute Error (MAE)	18,587.85
R-squared (R^2)	0.9862

The test-set R^2 of 0.9862 indicates that the model explains about 98.62% of the variation in the test data. However, this result should be interpreted cautiously: variables such as Discounts and COGS may be closely related to how Sales is calculated, so the score does not guarantee accurate future forecasting. Further validation and testing on genuinely future data are recommended.

5. Key Findings

- The dataset contains 700 sales records spanning 2013–2014.
- There are 53 missing values in Discount Band and 31 missing values in Date; no duplicate rows were found.

- Sales performance can be compared across five countries and six products.
- Profit can be compared across five business segments.
- Monthly and yearly sales visualizations provide a way to inspect trends over time.
- The regression model achieved MAE 18,587.85 and R^2 0.9862 on the selected test split, but additional validation is needed.

6. Recommendations

- Compare sales across countries and investigate the business factors behind differences.
- Review product-level sales to support inventory and sales planning.
- Monitor profit by segment, not sales alone, when assessing performance.
- Use monthly sales trends as an input to planning, while checking for missing dates.
- Investigate the missing Discount Band and Date values and document any imputation or exclusion.
- Validate the regression model on future or otherwise unseen data and review feature leakage before using it for forecasting.

7. Conclusion

This capstone combined exploratory analysis, a Linear Regression model, and a Power BI dashboard to summarize a sales dataset. The work demonstrates a basic end-to-end data-analysis workflow. Before using the model for operational forecasting, data-quality issues and predictive performance should be examined further.

8. Project Files

The project folder should include the original Excel dataset, week4_eda.py, week4_regression.py, week4_findings.py, and the saved Power BI report (week4 charts.pbix). Include this PDF as the written explanation report. Add dashboard screenshots if your submission instructions require them.